

Logan Chu

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EDUCATION

Duke University

Durham, NC

Bachelor of Science in Computer Science; GPA: 3.92

May 2028

- **Coursework:** ML Systems, Operating Systems, Computer Architecture, Distributed Systems, Computer Networks, Data Structures & Algorithms, Machine Learning & Deep Learning

SKILLS

Languages: Python, Rust, C++, C, Go, Java, Bash, SQL

ML Systems: PyTorch, FSDP, DDP, vLLM, SGLang, Quantization, KV Cache, Continuous Batching

GPU Systems: CUDA, Triton, Nsight Systems, Nsight Compute, Kernel Fusion, CUDA Graphs, torch.compile, XLA

Distributed Systems: NCCL, MPI, SLURM, Linux, AWS, GCP, RESTful APIs, Docker, Kubernetes, Bazel, Git, CI/CD

TECHNICAL EXPERIENCE

Yotta Labs

Dec 2025 – Jun 2026

Software Engineering Intern

- Built GPU performance analysis tooling using **Nsight Systems** and **Nsight Compute** to correlate hardware metrics with **PyTorch** operators, identify memory bottlenecks, and drive PyTorch graph optimizations, achieving **1.76–3.63×** kernel wall-time speedups across GPT-2 and SDPA workloads served through **vLLM** and **SGLang**.
- Developed a GPU profiler for a compute platform used by **50K+ developers**, including at MIT, Stanford, and Berkeley.

Amazon Web Services

Jun 2026 – Present

Software Development Engineering Intern

- Built a fault-tolerant Rust agent for Amazon's zero-trust access platform, automating cryptographic attestation workflows across developer environments.
- Designed **CI/CD** release infrastructure in AWS CDK, distributing the agent to thousands of developer machines.

Grill Lab

Apr 2025 – Sep 2025

Research Assistant

- Reduced segmentation model training time by **5×** through distributed training on the Duke Compute Cluster using **SLURM**.
- Built an MCP server integrated with Claude Desktop to automate **HPC** data management and training workflows, reducing manual job submission and workflow orchestration overhead.

Systems for Engaging the Environment Lab

Sep 2025 – Apr 2026

Research Assistant

- Built a geospatial **digital twin** pipeline in **NVIDIA Isaac Sim**, reconstructing trajectories from real-world GPS/IMU data with Extended Kalman Filtering, reducing noise **35%** and achieving **< 2m** error against ground truth.
- Trained and evaluated multivariate **time-series LSTM** forecasting and unsupervised **autoencoder** anomaly detection in **PyTorch**, cutting trajectory RMSE **18%** vs. a constant-velocity baseline and flagging corrupted GPS/IMU segments.

PROJECTS

Diffusion Transformer World Model for Robotic Manipulation

Jan 2026 – Apr 2026

- Reduced inference latency by **1.51×** ($79.6 \rightarrow 52.8$ ms/eval) by optimizing Flow Matching with a sliding-window KV cache, enabling real-time closed-loop control via receding-horizon **model predictive planning** with learned world models.
- Built an **8-worker parallel** data pipeline collecting 10k trajectories with failure-mode coverage.

Distributed Training

Oct 2025 – Jan 2026

- Implemented All-Reduce and All-to-All collectives from first principles using **MPI**, achieving correctness and stable scaling across 8 ranks with communication latency within ~ 10 –15% of MPI primitives for medium-sized tensors.
- Built tensor-parallel execution for Transformer FC layers with weight sharding and distributed forward/backward passes.

Fused GPU Inference Kernels

Oct 2025 – Feb 2026

- Implemented a custom Triton GPU kernel fusing GEMM, bias add, and ReLU with **shared-memory tiling** and FP16 register accumulation, delivering **30% runtime improvement** over PyTorch eager execution on T4 GPUs.
- Optimized GPU memory hierarchy and execution by eliminating intermediate writes via **operator fusion** and tuned tile sizes, **reducing global memory traffic** and kernel launches across 5,000-iteration latency-critical benchmarks.